

# Can Deterministic Network Verification Reduce Undetected Faults in LLM-Generated Configurations?

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CNSM 2026 Agentic AI Researcher Track

**Abstract**—We preregistered and executed a paired confirmatory study (study id c1-gnr-vv-2026-0001) to measure whether inserting a deterministic network verifier between an LLM intent→configuration generator and an emulated execution reduces the proportion of configurations that would execute with operational faults. Using shared\_initial\_candidate semantics (one identical LLM candidate per intent evaluated both without and with verification), we evaluated 72 preregistered paired intents with gpt-5-mini and a canonical deterministic verifier (deterministic\_netops\_validator\_v5). Execution completed with 144 episodes (72 pairs) and 72 model calls. All baseline executions produced no emulator-observed faults ( $n_{11} = 72$ ,  $n_{10} = n_{01} = n_{00} = 0$ ). The paired difference in undetected-fault proportion is 0.0 (paired bootstrap 95% CI [0.0, 0.0], exact McNemar two-sided  $p = 1.0$ ; bootstrap seed = 1729, resamples = 10000). Under the preregistered shared\_initial\_candidate contract this degenerate confirmatory outcome is expected when identical generated candidates produce no baseline execution faults. We report the preregistration rationale, deterministic execution accounting, representative validator diagnostics, exact inferential outputs, and artifact-grounded recommendations for follow-on confirmatory designs able to detect verifier utility under realistic baseline-error regimes.

## I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate high-level operator intents into device-level network configurations. This intent→configuration automation can speed change pipelines but risks silently violating reachability, access-control, ordering, or workflow invariants and thereby causing outages if generated text is executed without additional checks. A standard operational mitigation is a generate–verify–repair (GVR) architecture that combines stochastic LLM generation with deterministic verification and, if needed, repair or human review. However, despite many architectural proposals and prototypes, systematic preregistered experiments that measure a verifier’s detection performance against executed outcomes under tightly controlled paired designs are scarce.

This paper answers a narrowly scoped, preregistered research question: does inserting a deterministic network verifier between an LLM generator and emulator execution materially reduce the proportion of configurations that execute with operational faults when the identical generated candidate is evaluated both without and with verification? That estimand isolates verifier detection from improvements that could arise from re-sampling, prompt-engineering, or repair-enabled iterations. We executed study c1-gnr-vv-2026-0001

using gpt-5-mini and deterministic\_netops\_validator\_v5 across 72 preregistered paired intents under shared\_initial\_candidate semantics. The run completed with no technical failures, produced a degenerate confirmatory outcome (no baseline emulator faults), and yields a paired difference of 0.0 (95% CI [0.0,0.0], exact McNemar  $p = 1.0$ ). Rather than treating this as a null failure, we analyze why it occurred given the preregistered contract and archived artifacts, and we provide concrete, artifact-grounded guidance for designs that can detect verifier utility when baseline error prevalence is non-negligible.

## II. RELATED WORK

We position this study against three closely related strands: (1) LLM-based intent→configuration automation and emulated evaluations; (2) multi-agent and tool-augmented NetOps frameworks that propose verifier integration; and (3) generate–verify–repair and constrained-decoding methods from software/code domains that inform possible NetOps adaptations.

LLM intent→configuration translation. Recent empirical work demonstrates that instruction-tuned LLMs can map operator intents to device-level configuration text within constrained vendor dialects and curated testbeds. For example, Nguyen et al. present an intent→configuration pipeline and evaluate LLM performance on structured configuration tasks, documenting promising task-level accuracy but highlighting the gap between textual correctness and safe execution in operational environments [1]. Those studies use emulator or synthesized workloads to assess functional correctness of generated artifacts but emphasize that textual matching metrics do not substitute for execution-level invariants such as reachability or ordering constraints. Our study adopts the same operational framing (emulator execution as a conservative proxy for operational fault) but differs methodologically by preregistering a paired detection estimand under shared\_initial\_candidate semantics to isolate verifier detection capability on identical generated candidates.

Multi-agent and verifier-integrating NetOps systems. Several prototypes and architectural reports advocate tool-augmented or multi-agent LLM systems that incorporate verification steps to reduce regressions and to enable safe automation at scale. Notably, recent multi-agent intent-driven frameworks demonstrate how separate agent roles (planner, config generator, validator) can be orchestrated to produce and check configurations before execution; these systems show

how verifier signals can be folded into human-in-the-loop or automated repair workflows but typically report system behavior at the demonstration or prototype level rather than in preregistered paired detection trials [2]. Our experiment operationalizes a single, canonical verifier in a strict generate–verify evaluation contract to measure whether the verifier can, by itself and without changing the generated candidate, prevent executed faults that would otherwise occur.

Generate–Verify–Repair and constrained decoding proveance. The generate–verify–repair (GVR) pattern has a substantial methodological pedigree in program repair and structured-output generation. Architectural proposals and empirical studies in code generation and self-repair show that deterministic checkers (unit tests, static analyzers) combined with stochastic generation can enable iterative repair loops that converge on functionally correct outputs [3]. Complementary work on constrained decoding and integer-programming-based decoding demonstrates that enforcing structural constraints during generation reduces syntactic or semantic violations in structured domains [4]. These results motivate two dimensions of adaptation for NetOps: (a) constraining or biasing generation toward verifier-satisfiable forms (constrained decoding), and (b) allowing verifier-triggered repair iterations that change the executed candidate distribution. Importantly, those adaptations change the causal estimand: they do not measure verifier detection on an identical candidate but rather the end-to-end benefit of verifier-enabled generation and repair. The preregistered `shared_initial_candidate` semantics used here intentionally hold the generated candidate fixed across conditions so that any paired difference can be attributed only to verifier detection (not to sampling, constrained decoding, or repair).

Deterministic network verification and policy checking. Deterministic analysis tools developed in the networking literature (header-space analysis and related formal checkers) provide precise, semantics-aware invariants for reachability, ACL, and forwarding behavior; these tools are natural verifier candidates for LLM-generated configs because they can conservatively flag violations that imply operational harm [5]. Prior work on real-time policy checking and emulated validation shows how formal analyses can be integrated into testbeds to prevent regressions before deployment. Our canonical verifier (`deterministic_netops_validator_v5`) was used in the same role: apply deterministic, rule-based checks that compare the generated sequence against manifested workflow constraints (sequence, allowed fields, and protected interfaces) encoded in each task manifest. Where prior literature argues for verifier integration at the architecture level, our contribution is a preregistered paired measurement that quantifies verifier detection power when the generated candidate is held identical across conditions.

Synthesis and gap. Collectively the literature indicates (a) LLMs can produce plausible device commands under constrained settings, (b) multi-agent and tool-augmented systems often propose a verifier role, and (c) GVR and constrained decoding strategies can materially change generation out-

comes by construction. What is missing—and what this study targets—is a preregistered, paired, artifact-grounded measurement of verifier detection performance on identical generated candidates: that specific estimand isolates the marginal detection capacity of a verifier decoupled from generation modifications. The degenerate result we observe (zero baseline emulator faults) is thus directly informative: it reveals a design regime—highly constrained synthetic intents and generator behavior—where detection cannot be measured because baseline faults are absent. We use archived traces and validator diagnostics to show how task specification and verifier rule alignment created that regime, and we suggest preregistered alternatives (independent sample arms, repair flows, adversarial task banks) that the literature suggests would expose measurable verifier utility when baseline error prevalence is non-negligible.

### III. METHODOLOGY

Preregistration and primary estimand. We preregistered and froze study `c1-gnr-vv-2026-0001` before observing outcomes. The primary estimand is the paired reduction in undetected operational-fault proportion (`paired_success_rate_difference_guarded_minus_baseline`). The confirmatory hypothesis (H1) targeted an absolute paired reduction of 0.20 (two-sided  $\alpha = 0.05$ , power  $\geq 0.80$ ); a sample-size heuristic yielded  $\approx 63$  pairs and we preregistered  $N = 72$  pairs (24 per topology-difficulty stratum). The preregistration and analysis plans, including sensitivity analyses and exclusion rules, are part of the archived preregistration record.

`Shared_initial_candidate` semantics and experimental contract. The `execution_contract` enforced `shared_initial_candidate` semantics: for each intent we performed exactly one initial LLM generation and evaluated that identical candidate in two paired conditions. Baseline condition: execute the shared candidate in an isolated emulator and record whether emulator-observed operational faults occur. Guarded condition: run the canonical deterministic verifier on the same candidate; if the verifier flags violations the guarded outcome is recorded as prevented (no executed fault); if the verifier does not flag, execute the identical candidate in the emulator and record the result. By construction, any observed paired difference can only arise from the verifier preventing execution of a candidate that would otherwise have failed at emulation; this isolates detector utility from generator-side resampling or repair effects.

Model, adapter, and verifier configuration (artifact-grounded). The declared model is `gpt-5-mini` (`execution/model_configuration.json`), requested via the `hosted_netops_gvr_v1` adapter. Key recorded model parameters: `declared_model_version = "gpt-5-mini"`, `provider = "openai_responses"`, `max_output_tokens = 2000`, `maximum_attempts_per_call = 1`, and `temperature = null`. The recorded `temperature = null` indicates that provider-default runtime sampling semantics were used; we treat this as an archived sampling-parameter uncertainty rather

than an explicit numeric temperature. The canonical verifier is `deterministic_netops_validator_v5` (`validator_version = "5.0"`) configured to run the `full_intent_constraint_validation` rule set. The verifier emits structured `validator_traces` per episode (fields include `parsed_command_count`, `required_command_count`, `normalized_configuration`, `validation_before` and `validation_after` final\_state snapshots, `violation_codes`, `violation_count`, and difficulty metadata). The verifier’s validity verdict and trace for each episode were archived under `execution/scoring/` and formed the mechanistic basis for guarded decisions.

Task-bank construction, contamination controls, and stratification. Confirmatory tasks were generated deterministically by `deterministic_netops_task_generator_v5`. Each `task_manifest` entry encodes: a natural-language intent, a machine-structured `required_sequence` (ordered field changes that implement the intent), `allowed_touched_fields`, preserved-state policies, initial and expected\_final\_state snapshots, and a difficulty.level tag (`medium/hard/very_hard`). The preregistration required deterministic exact-match contamination checks against a public-corpus fingerprint (`analysis/contamination_summary.csv`); exact-match flagged items are excluded from confirmatory analysis (none were excluded in this run). The confirmatory sample followed the preregistered stratification (24 pairs per difficulty stratum) to allow descriptive subgroup inspection.

Execution protocol, retries, and deterministic accounting. The `missingness_plan` permitted up to two retries (three attempts total) for transient hosted-API errors on generation calls; all retries and provider events were logged. The run started at 2026-08-31T06:56:23.065707Z and completed at 2026-08-31T07:06:28.665518Z (execution manifest). The execution manifest records `planned_episode_count = 144`, `completed_episode_count = 144`, `failed_episode_count = 0`, and `model_calls_used = 72` (one shared generation per pair). Provider-call audit (`deterministic_reconciliation.json`) reports `hosted_model_call_event_count = 72`, `completed_hosted_model_call_event_count = 72`, `failed_hosted_model_call_event_count = 0`, `cache_miss_count = 72`, and `stage_counts = {"shared_initial_generation": 72}`. These deterministic accounting artifacts were used to reconcile paired outcomes and to confirm that the `shared_initial` generation stage was the only cross-condition generation activity.

Scoring, validation rules, and per-episode traces. For each episode the verifier produced a binary validity verdict (`valid / invalid`) and an accompanying `validator_trace` JSON that documents `parsed` and `required` command counts, `normalized_configuration`, `validation_before/after` states, and any `violation_codes`. The scoring rule deployed was `full_intent_constraint_validation` (validator enforces `required_sequence` order, `allowed_touched_fields`, and `preserve_unspecified_state` constraints encoded in the task manifest). Per-episode scoring artifacts (`execution/scoring/task-*.json`) therefore permit audit of why a candidate was accepted or flagged. In the recorded

run no verifier flagging occurred for confirmatory pairs (`violation_count = 0` across representative episodes), and no automated repair was applied (`repair_applied = false` in `validator_trace` fields).

Inferential and sensitivity analysis procedures (deterministic scripts). The primary analysis used the registered `paired_binary_analysis_v1` executor to construct the  $2 \times 2$  contingency table (`guarded \times baseline`) using binary indicators where an undetected executed fault is 1 and success/no-fault is 0. The prespecified exact inferential test is an exact McNemar two-sided test when discordant pairs exist. We also computed paired bootstrap-percentile 95% confidence intervals for the paired difference using 10,000 resamples (`bootstrap_seed = 1729`), recorded in `analysis/analysis_log.jsonl`. Pre-specified subgroup confirmatory comparisons (topology complexity and policy type) were registered with Holm correction to control familywise error; these controls are implemented in the deterministic analysis pipeline but become non-informative in the absence of discordant pairs. Pre-registered sensitivity analyses executed by the same scripts include (a) treating excluded confirmatory pairs as failures (`complete_pair_primary_with_failure_as_zero_sensitivity`) and (b) bootstrap diagnostics for detector detection and false-positive rates. Because the run produced no exclusions and no discordant outcomes these sensitivity analyses reproduce the degenerate numerical conclusion.

Design-level notes and construct tradeoffs (artifact-linked). The `shared_initial_candidate` semantics intentionally isolates verifier detection from generator-side sampling; this preserves a clean causal interpretation of verifier-prevention utility but prevents measurements that rely on verifier-triggered resampling or repair. The archived `execution/model_configuration.json` recording `temperature = null` documents that provider-default sampling behavior was allowed; had we required explicit sampling parameters (non-null temperatures) we could have measured sampling-sensitivity as a separate factor. The verifier’s `full_intent_constraint_validation` rule set enforces task-manifest constraints: because the task generator encoded explicit `required_sequence` and `allowed_touched_fields` many generated candidates closely matched those constraints, which the verifier therefore did not flag. This alignment is measurable in `validator_trace` fields (`parsed_command_count` equals `required_command_count` and `violation_count = 0` for representative episodes) and explains the ceiling result when baseline emulator faults are absent. The preregistered sensitivity and alternative contracts (independent-generation arms, repair-enabled flows, adversarial task seeding, and multi-model comparisons) are described in the preregistration and can be executed using the same deterministic execution and analysis infrastructure archived with this run.

#### IV. RESULTS

Execution and manifest summary. The autonomous pipeline completed without technical failures and with full deterministic accounting. Key run-level figures from

the archived manifests are: `planned_episode_count = 144`; `completed_episode_count = 144`; `failed_episode_count = 0`; `model_calls_used = 72`; `artifact_count = 510`. The run-level timestamps (`started_at_utc = 2026-08-31T06:56:23.065707+00:00`; `completed_at_utc = 2026-08-31T07:06:28.665518+00:00`) indicate a wall-clock duration of  $\approx 10.1$  minutes. `analysis/condition_summary.csv` (archived) records baseline successes = 72, baseline failures = 0 (`success_rate = 1.0`) and guarded successes = 72, guarded failures = 0 (`success_rate = 1.0`).

Paired contingency, inferential outputs, and bootstrap diagnostics. Aggregating across the 72 confirmatory pairs produced the exact paired contingency counts `n_11 = 72`, `n_10 = 0`, `n_01 = 0`, `n_00 = 0` (guarded  $\times$  baseline). The primary estimand `paired_success_rate_difference_guarded_minus_baseline = 0.0`. The preregistered exact McNemar two-sided test produces `p = 1.0` given the absence of discordant pairs. The paired bootstrap-percentile 95% confidence interval for the paired difference (`bootstrap_seed = 1729`; `bootstrap_resamples = 10000`; `analysis/analysis_log.jsonl`) is `[0.0, 0.0]`, reflecting the degenerate sampling distribution when all paired outcomes are concordant. `analysis/paired_contingency_table.csv` and `analysis/condition_summary.csv` contain these tabulations and are archived.

Per-episode validator diagnostics and representative exemplars. We inspected archived per-episode scoring JSONs under `execution/scoring/` and representative task-manifest entries under `manuscript_evidence`. Across the sampled exemplars the verifier produced no violation flags. Representative excerpts from archived scoring artifacts: - `pair-000001` (`task-000001`): `difficulty.level = "medium"`; `parsed_command_count = 4`; `required_command_count = 4`; `validator_version = "5.0"`; `violation_count = 0`; `normalized_configuration` equals the shared candidate (`execution/scoring/task-000001-*.json`). - `pair-000002` (`task-000002`): `difficulty.level = "hard"`; `parsed_command_count = 2`; `required_command_count = 2`; `violation_count = 0`. - `pair-000003` (`task-000003`): `difficulty.level = "hard"`; `parsed_command_count = 6`; `required_command_count = 6`; `violation_count = 0`. For additional archived representative tasks the task manifests encode `required_command_count` and `difficulty` tags (examples: `task-000004` `required_command_count = 4`, `difficulty.level = "very_hard"`; `task-000005` `required_command_count = 3`, `difficulty.level = "very_hard"`; `task-000006` `required_command_count = 6`, `difficulty.level = "very_hard"`). In all archived scoring JSONs for confirmatory pairs the `verifier_traces` report `violation_count = 0` and `validation_after.valid = true`, demonstrating that the verifier processed each shared candidate and did not flag the candidate as violating the manifest-encoded constraints.

Contamination, missingness, and sensitivity analyses. `analysis/contamination_summary.csv` records zero exact-match contamination flags for confirmatory pairs. `analysis/missingness_summary.csv` records `complete_pairs = 72` and zeros for failed or unscorable episodes. The

preregistered sensitivity analyses (including the conservative treatment that would count excluded pairs as failures and bootstrap diagnostics) were executed by the registered analysis executor; because there were no exclusions and no discordant pairs these sensitivity analyses reproduce the degenerate numeric conclusion (`paired_difference = 0.0`). All analysis artifacts (`analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl`) are archived and reference the bootstrap seed and resample count used.

Deterministic reconciliation and provider-call audit. The `deterministic_reconciliation.json` confirms accounting consistency: `completed_episode_count = 144`, `complete_pair_count = 72`, `baseline_success_count = 72`, `guarded_success_count = 72`, `n_11 = 72`, `n_10 = 0`, `n_01 = 0`, `n_00 = 0`, and `all_deterministic_consistency_checks_passed = true`. Provider-call audit fields show `hosted_model_call_event_count = 72`, `completed_hosted_model_call_event_count = 72`, `failed_hosted_model_call_event_count = 0`, `cache_miss_count = 72`, and `stage_counts = {shared_initial_generation: 72}`, consistent with the preregistered `shared_initial_candidate` semantics and the execution manifest.

Interpretation of the degenerate outcome (evidence-grounded). The complete absence of baseline emulator faults in the archived confirmatory corpus is the proximate cause of the degenerate paired result. The archived artifacts together document measurable contributors: (1) the deterministic task generator produced constrained intents with manifest-encoded `required_sequence` and `allowed_touched_fields` that align the generation objective tightly with verifier checks; (2) `gpt-5-mini` (declared in `execution/model_configuration.json`) produced configurations that matched those manifest constraints across the sampled tasks; and (3) the `deterministic_netops_validator_v5` rule set (`full_intent_constraint_validation`) enforces the same manifest-encoded sequence and field constraints, producing concordant validity judgments. These archived, machine-verifiable facts explain why the verifier could not reduce executed faults under the `shared_initial_candidate` contract: there were no executed faults to detect.

## V. DISCUSSION

Confirmatory interpretation and boundary conditions. The preregistered paired design with `shared_initial_candidate` semantics validly measures verifier detection on identical candidates; because the identical generated candidates produced zero emulator faults the verified paired outcome is a ceiling/null result: the verifier could not reduce executed faults that did not occur. The numeric outputs (`n_11 = 72`; `paired_difference = 0.0`; McNemar `p = 1.0`; bootstrap CI `[0.0, 0.0]`) are direct consequences of the preregistered contract and observed data and therefore correctly reported as a degenerate confirmatory result.

Artifact-grounded causes of the ceiling outcome. Archived artifacts allow us to identify measurable factors that explain the ceiling outcome. Three concrete, artifact-supported points

stand out: 1) Task-manifest constraint alignment. The deterministic task generator encoded explicit `required_sequence` fields, `allowed_touched_fields`, and `workflow_policies` for every confirmatory item (see representative `task_payload` entries under `manuscript_evidence_bundle/representative_tasks`). Validator traces repeatedly show `parsed_command_count == required_command_count` for representative episodes (e.g., task-000001: `parsed=4 required=4`; task-000003: `parsed=6 required=6`), indicating the generated candidate matched the structured sequence encoded in the task manifest. When generation and manifest constraints are strongly congruent, room for verifier-detectable deviations shrinks. 2) Model-task congruence in a synthetic, constrained bank. The declared model (`execution/model_configuration.json: declared_model_version = "gpt-5-mini"`) produced candidates that matched manifest constraints across the synthetic corpus. Because `execution/model_configuration.json` records `temperature = null`, the run used provider-default sampling semantics; the archived provider-call audit shows `hosted_model_call_event_count = 72` with `cache_miss_count = 72`, indicating fresh calls for each shared generation. Under provider-default sampling and the deterministic, high-clarity prompts used by the generator, the model produced stable, manifest-conformant outputs for the sampled items. 3) Verifier-manifest semantic overlap. The canonical verifier's rule set (`deterministic_netops_validator_v5_full_intent_constraint_validation`) enforces the same sequence and allowed-field constraints encoded by the task generator. This semantic overlap is visible in per-episode `validation_before/after` snapshots archived under `execution/scoring/*`.json: validator traces show both `final_state` snapshots matching `expected_final_state` and `violation_count = 0` across examples. When verifier checks are essentially restatements of the task manifest constraints, and generators respect those constraints, the verifier has limited opportunity to detect faults.

Statistical and design implications. From a statistical point of view the paired estimand requires discordant pairs ( $n_{10}$  or  $n_{01} > 0$ ) to infer detection difference. A zero-prevalence baseline (no executed faults) collapses the paired analysis: the permutation of outcomes offers no information about verifier utility. This outcome does not imply the verifier would always be redundant; it indicates that, for this synthetic bank and sampling semantics, baseline error prevalence was effectively zero. The preregistered power calculation (target paired absolute reduction 0.20) assumed nonzero baseline prevalence; that assumption is violated here and the power argument no longer applies.

What these artifacts imply for follow-on confirmatory designs. The archived evidence suggests concrete, preregisterable modifications that would produce informative, non-degenerate paired comparisons while preserving rigorous controls: - Independent-generation arm: register independent generator draws per condition (baseline vs guarded) so verifier-mediated repairs or sampling variability can change executed-candidate distributions. This removes the `shared_initial_candidate` constraint and allows measuring downstream benefits of verifica-

tion+repair or alternative sampling strategies. - Repair-enabled flows: permit a single verifier-triggered repair call (as allowed by the `execution_contract`'s `transformation_scope` options) and measure whether repaired candidates reduce emulator faults compared to un-repaired executions. The artifact traces already record `repair_applied` flags and `repair_prompt_sha256` placeholders for episodes where repair would be invoked. - Adversarial or ambiguity-seeded task bank: include items with intentionally underspecified intents, vendor-edge cases, or ordering subtleties where `required_sequence` is ambiguous. The deterministic task generator's payload structure supports seeding such ambiguities while preserving provenance checks; `analysis/contamination_summary.csv` shows the existing pipeline can detect exact-match contamination and thus preserve pre-registration integrity for adversarial items. - Explicit sampling parameterization: set non-null temperature and record it in `execution/model_configuration.json` to quantify sampling-driven variability. The current `temperature = null` is a residual uncertainty; making sampling explicit will allow controlled exploration of model stochasticity and quantification of verifier value across sampling regimes. - Multi-model comparisons: include multiple model versions or vendors to measure verifier utility across generators. The `deterministic_reconciliation.json` shows the adapter supports logging provider-call audits per model and would enable stratified inference across generators.

Operational tradeoffs and measurable verifier metrics. The artifact vocabulary supports more granular operational metrics than a binary undetected-fault indicator: verifier true-positive rate (proportion of emulator failures flagged), verifier false-positive rate (proportion of verifier flags that would not produce emulator failures), blocking cost (proportion of verifier flags that prevent benign changes), and repair success rate (proportion of repaired candidates that execute without faults). `analysis/paired_contingency_table.csv` and `analysis/condition_summary.csv` illustrate how a richer contingency scheme can be recorded. Measuring blocking cost requires permitting verifier-blocked candidates to be inspected or replayed under a repair-enabled flow; the current guarded protocol records prevented outcomes but does not execute prevented candidates, so blocking-cost estimates are not available from this run.

Threats to validity revisited with artifact evidence. Internal validity: the preregistered `shared_initial_candidate` semantics intentionally isolates verifier detection but prevents any verifier-influenced change in executed candidate distributions; this choice explains why a degenerate result may occur in low-error baselines. Construct validity: the emulator-observed fault label is a conservative operational proxy, but emulator-level tests omit traffic-dependent and hardware-specific failure modes; validator traces record `final_state` snapshots but cannot capture live-traffic interaction effects. External validity: experiments used a single declared model (`gpt-5-mini`), a synthetic task bank created by `deterministic_netops_task_generator_v5`, and a single deterministic verifier; generalization to other generators, vendor dialects, or production hardware is limited. Conclusion validity: with baseline failure prevalence = 0

the confirmatory paired test lacks the necessary discordance to make inferential claims about verifier usefulness under different baseline regimes.

Practical recommendations grounded in the archive. Based on the archived manifests and traces we recommend that future preregistered studies aiming to quantify verifier utility should: 1) explicitly select or seed a baseline failure prevalence (via adversarial items, ambiguous intents, or historical change examples) so the paired estimand has scope to detect reductions; 2) preregister whether `shared_initial_candidate` semantics or independent-generation semantics will be used, because they answer different operational questions (detection on identical outputs vs. end-to-end improvement under verifier-mediated change); 3) record explicit sampling parameters (non-null temperature) in `execution/model_configuration.json` to remove provider-default ambiguity; 4) include repair-enabled arms to measure both detection and net operational improvement; and 5) log verifier-blocking events and, where ethically permissible, execute or simulate blocked candidates to estimate blocking cost. All these recommendations map directly to fields and artifacts present in the current evidence bundle (task manifests, `execution/model_configuration.json`, `execution/scoring/*.json`, and analysis logs) and can be implemented without inventing new tooling.

Concluding perspective. The degenerate confirmatory outcome reported here is an empirically supported datapoint: under the tested constrained, synthetic conditions the generator and manifest constraints produced valid candidates and a canonical verifier did not change executed outcomes. The key scientific value of the run is therefore methodological and procedural: the archived artifacts demonstrate a fully preregistered, deterministic pipeline for measuring verifier detection under tightly controlled semantics, and they make explicit which preregistration choices (`shared_initial_candidate` vs independent generation, sampling parameterization, adversarial seeding, repair permissions) determine whether future confirmatory tests can meaningfully quantify verifier utility.

## VI. LIMITATIONS

- Confirmatory ceiling/null outcome: the baseline generator produced zero emulator faults on the preregistered task set, preventing direct measurement of verifier detection benefit.
- `Shared_initial_candidate` semantics: the design measures verifier effect on the same generated candidate only and does not assess independent per-condition generation, multi-sample generation, or repairs unless explicitly permitted by the contract.
- Single-model scope: experiments used one declared model (gpt-5-mini); results do not imply generality across models or versions.
- Synthetic/emulated tasks: task corpus and emulator executions differ from production devices and live traffic; external validity to production deployments is limited.
- Sampling-parameter uncertainty: `execution/model_configuration.json` shows `temperature = null`

(sampling parameters unset); null indicates provider-default runtime sampling semantics and is a residual uncertainty recorded in the archived configuration.

- Residual contamination risk: deterministic provenance and exact-match checks were applied, but private training-data contamination of the hosted model cannot be fully ruled out and remains residual uncertainty.

## VII. CONCLUSION

We executed a preregistered paired evaluation (c1-gnr-vv-2026-0001) under `shared_initial_candidate` semantics to test whether a canonical deterministic verifier reduces the proportion of LLM-generated configurations that would execute with operational faults. For the chosen model (gpt-5-mini), verifier (`deterministic_netops_validator_v5`), and synthetic/emulated task bank, baseline executions produced zero emulator faults and the verifier did not change outcomes (paired difference = 0.0; bootstrap 95% CI [0.0,0.0]; exact McNemar  $p = 1.0$ ). This artifact-grounded null/ceiling outcome is internally consistent with the preregistered design: when identical generated candidates produce no executed faults a verifier cannot reduce executed faults under the `shared_initial_candidate` contract. Measuring verifier utility under realistic error regimes requires preregistered contracts that permit independent or multiple generator samples, repair-enabled flows, adversarial task sets, and multi-model comparisons. The archived artifacts (responses, task manifests, validator traces, execution manifest, and analysis logs) provide a reproducible baseline and a template for those follow-on confirmatory designs and power calculations.

## DISCLOSURE STATEMENT

Autonomy and artifacts: This study was executed autonomously after an autonomy lock. Human role: a Research Director selected the topic family and approved the immutable initial master prompt prior to the autonomy lock; no human manually edited technical manuscript content after the lock. Models and tools: initial generation used gpt-5-mini (declared `model_version` recorded in `execution/model_configuration.json`) orchestrated via the `hosted_netops_gvr_v1` adapter; `deterministic_netops_validator_v5` (`validator_version = "5.0"`) served as the canonical verifier; an isolated emulator executed candidate configurations. Preregistration: study c1-gnr-vv-2026-0001 was preregistered and frozen before observing outcomes. Immutable master prompt reference: `provenance/master_prompt.txt` (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). Key archived artifacts include `execution/raw_results.jsonl`, `execution/task_manifest.jsonl`, `execution/model_configuration.json`, `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `deterministic_reconciliation.json`, and per-episode validator traces under `execution/scoring/`. The model configuration records `temperature = null` (provider-default sampling semantics archived in

execution/model\_configuration.json). Provider-call audit: hosted\_model\_call\_event\_count = 72. No human manual manuscript editing or content insertion occurred after the autonomy lock.

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